

Stochastic Gradient Descent

Optimization Principles

Hoa T. Vu

Vector Notation

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Gradient $\nabla_{\theta} \mathcal{L}(\theta) = \frac{-2}{n} \sum_{i=1}^n (y_i - \theta^\top x_i) x_i$

Motivation

Full Gradient Descent

Each update requires the gradient over *all* n samples:

$$\nabla \mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^n \nabla \ell_i(\theta)$$

where $\ell_i(\theta)$ is the loss on the i -th sample alone.

Every parameter update touches every data point.

MSE: Each Term Requires One Sample

Per-sample gradient for linear regression $\hat{y}_i = \theta^\top x_i$:

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The full gradient averages over all n samples:

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Why This Is Expensive

- One gradient step = n individual gradient evaluations
- For $n = 10^6$, that is 10^6 dot products per update
- Modern datasets have $n = 10^8$ or more

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Key Question

Can we speed up gradient descent?

Example: Full Gradient for MSE

Model $\hat{y}_i = \theta x_i$, loss $\ell_i(\theta) = (y_i - \theta x_i)^2$, gradient $\nabla_{\theta} \ell_i = -2(y_i - \theta x_i) x_i$.

i	x_i	y_i	$\hat{y}_i$ ($\theta = 2$)	$\nabla_{\theta} \ell_i$
1	1	3	2	$-2(3 - 2)(1) = -2$
2	2	5	4	$-2(5 - 4)(2) = -4$
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With $n = 10^6$ rows, every single update requires 10^6 such computations.

SGD

Stochastic Gradient Descent

At each step, pick **one** sample i_t uniformly at random:

$$\mathbf{g}_t = \nabla \ell_{i_t}(\theta_t), \quad \theta_{t+1} = \theta_t - \alpha_t \mathbf{g}_t$$

Cost per step: **one** gradient evaluation instead of n .

SGD: Unbiased Estimate

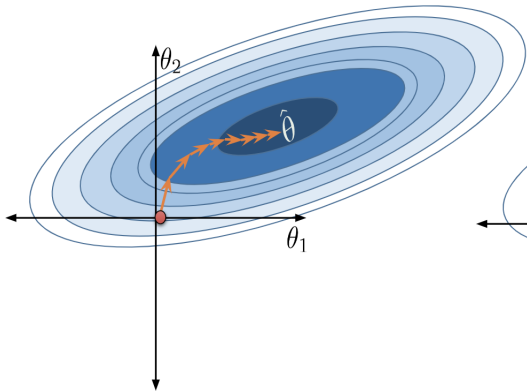
The single-sample gradient g_t is a noisy but **unbiased** estimate:

$$\mathbb{E}[g_t] = \nabla \mathcal{L}(\theta_t)$$

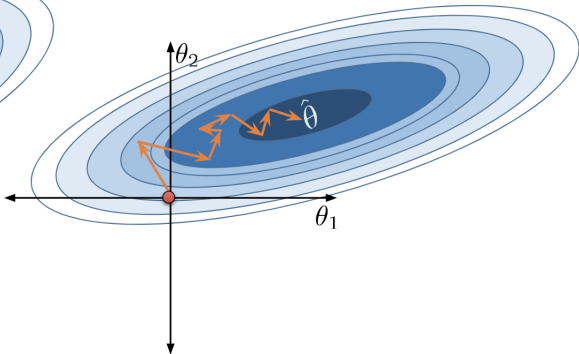
On average, g_t points in the same direction as the true gradient.

Visualization

Batch Gradient Descent



Stochastic Gradient Descent



SGD: Pros and Cons

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- One pass per step (cheap)
- Can escape shallow local minima
- Works in online / streaming settings

Cons

- High variance, erratic path
- Requires decreasing α_t to converge
- Harder to parallelize than mini-batch

Mini-batch SGD

Use a random subset $\mathcal{B}_t \subseteq \{1, \dots, n\}$ of size B :

$$g_t = \frac{1}{B} \sum_{i \in \mathcal{B}_t} \nabla \ell_i(\theta_t), \quad \theta_{t+1} = \theta_t - \alpha_t g_t$$

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Cost/step	cheap	moderate	expensive
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In practice $B \in \{32, 64, 128, 256\}$ balances noise and hardware efficiency.

Implementation

SGD in PyTorch

```
from torch.utils.data import (  
    TensorDataset, DataLoader)  
  
loader = DataLoader(  
    TensorDataset(X, y),  
    batch_size=32, shuffle=True)  
  
opt = torch.optim.SGD(  
    model.parameters(), lr=1e-3)  
  
for epoch in range(num_epochs):  
    for X_b, y_b in loader:  
        loss = criterion(model(X_b), y_b)  
        opt.zero_grad()  
        loss.backward()  
        opt.step()
```

- DataLoader shuffles and batches automatically
- One full pass = one **epoch**
- `opt.step()` uses the mini-batch gradient only

Full GD

1. For each epoch:
2. Compute gradient on *all* data
3. Update θ

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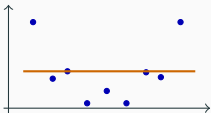
With n samples and batch size B , one epoch performs $\lceil n/B \rceil$ gradient steps instead of one.

Bias-Variance Tradeoff

Underfitting, Overfitting, and Just Right

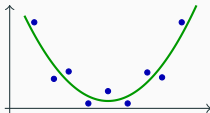
Underfitting

high bias, low variance



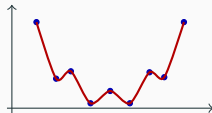
Just Right

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Overfitting

low bias, high variance



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Bias how far off the average prediction is from the truth

Variance how much the prediction changes across datasets

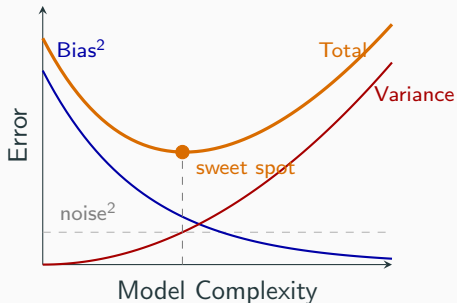
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$$\mathbb{E}[\text{error}] = \underbrace{\text{Bias}^2}_{\text{underfitting}} + \underbrace{\text{Variance}}_{\text{overfitting}} + \sigma^2$$

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The **sweet spot** is the minimum of the total error curve.



Regularization

What Is Regularization?

Add a penalty to the loss to discourage large weights:

$$\mathcal{L}_{\text{reg}}(\theta) = \underbrace{\mathcal{L}(\theta)}_{\text{fit the data}} + \lambda \underbrace{R(\theta)}_{\text{penalty}}$$

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L2 (Ridge) $R(\theta) = \|\theta\|_2^2$ $\rightarrow$ shrinks all weights, but never to exactly zero

L1 (Lasso) $R(\theta) = \|\theta\|_1$ $\rightarrow$ drives some weights to **exactly zero**

Subgradient Descent for L1

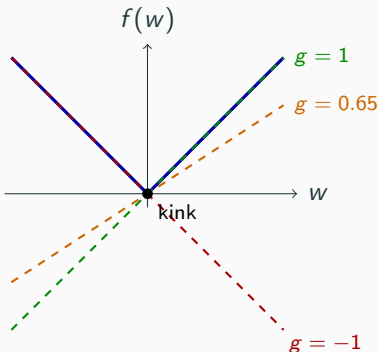
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A slope g is valid if its line lies *below* f everywhere.



Subgradient of $|w|$: Using $\text{sign}(w)$

Gradient descent needs $\frac{d}{dw}|w|$ at every step — but that derivative does not exist at $w = 0$.

Fix: pick any subgradient — the standard choice is $\text{sign}(w)$:

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At $w = 0$ the valid range is $[-1, +1]$; convention picks $g = 0$ — already at the minimum.

SGD with L1 Regularization

Replace the L1 gradient with $\text{sign}(\theta)$ and run SGD as usual:

$$\theta_{t+1} = \theta_t - \alpha \left(\nabla \ell_{i_t}(\theta_t) + \lambda \text{sign}(\theta_t) \right)$$

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Each step nudges every weight toward zero by $\alpha\lambda$; if a weight would cross zero, it snaps to zero and stays there.

Problem Set

Problem 1: Cross-Entropy Loss

Binary cross-entropy for true label $y \in \{0, 1\}$ and predicted probability $\hat{y} = \sigma(\theta^\top x)$:

$$\ell(\theta) = -y \log \hat{y} - (1 - y) \log(1 - \hat{y})$$

1. **Behavior.** What happens to ℓ as $\hat{y} \rightarrow 0$ and $\hat{y} \rightarrow 1$ when $y = 1$? Repeat for $y = 0$. Sketch the loss as a function of $\hat{y}$ for each case.
2. **Gradient.** Compute $\nabla_{\theta} \ell(\theta)$.
Hint: chain rule, and $\sigma'(z) = \sigma(z)(1 - \sigma(z))$. Simplify as far as possible.

Problem 2: Subgradients

(a) ReLU: $f(x) = \max(0, x)$

- Write down $\partial f(x)$ for all $x \in \mathbb{R}$.
- What value is conventionally used at $x = 0$ in practice, and why?

(b) SReLU: $f(x) = (\text{clip}(x, 0, 1))^2$, $\text{clip}(x, a, b) = \max(a, \min(b, x))$

- Write down $f(x)$ as a piecewise function.
- Write down $\partial f(x)$ for all x . Identify every point of non-differentiability and give the full subdifferential there.

Problem 3: Implementing SGD

t	y	t	y
-3.00	-76.39	0.43	-3.77
-2.57	-55.41	0.86	-3.61
-2.14	-34.20	1.29	3.21
-1.71	-21.28	1.71	8.84
-1.29	-18.66	2.14	16.65
-0.86	-12.17	2.57	32.36
-0.43	-6.37	3.00	49.94
0.00	-5.63		

Fit $\hat{y} = at^3 + bt^2 + ct + d$, MSE loss,
 $\theta_0 = (0, 0, 0, 0)$.

1. **From scratch.** Implement SGD in NumPy: shuffle each epoch, one sample per update. Run 200 epochs, $\alpha = 10^{-4}$. Report (a, b, c, d) and plot the loss curve.
2. **PyTorch.** Re-implement with `torch.optim.SGD`. Confirm you reach the same parameters.
3. (*Bonus*) Add L1 ($\lambda = 0.01$) using $\text{sign}(\theta)$ from Problem 2. Which coefficient shrinks first, and why?

Problem 4: True or False

For each statement, say whether it is **true or false** and give a one-sentence justification.

1. The SGD gradient estimate is biased: on average it points in a different direction than the true gradient.
2. Using a larger mini-batch always leads to a better final model.
3. A model with zero training loss will always have zero test loss.
4. L1 regularization can set weights to exactly zero; L2 regularization cannot.
5. If $\alpha_t = 0.1$ for all t , SGD is guaranteed to converge to the global minimum of a convex loss.